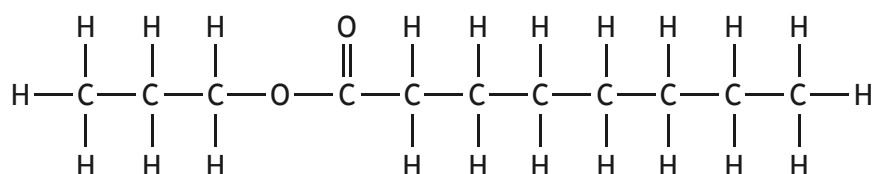


4. Coconut oil contains a mixture of compounds.

(a) Propyl octanoate is a compound found in coconut oil.



(i) Name the functional group in propyl octanoate.

1

(ii) Draw a structural formula for the carboxylic acid formed by hydrolysis of propyl octanoate.

1

(iii) An isomer of propyl octanoate with the same functional group was hydrolysed. One of the products of this hydrolysis was butanoic acid. Suggest a name for the other product.

1

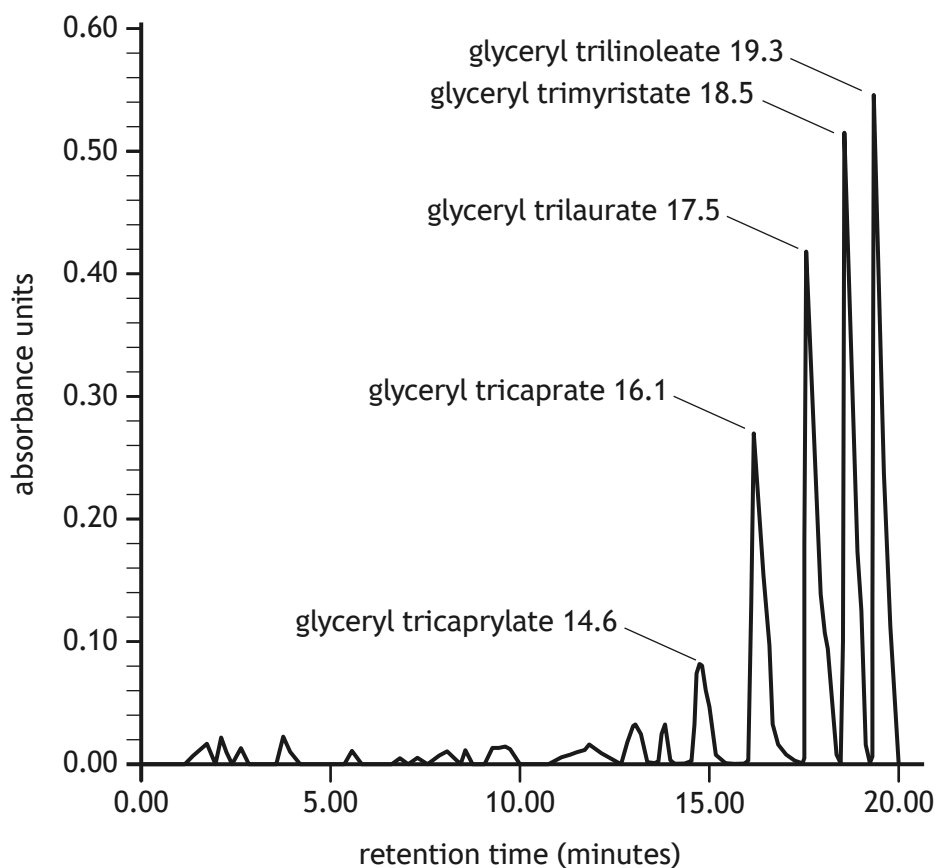
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* X 8 1 3 7 6 0 1 0 9 *

4. (continued)

- (b) Chromatography can be used to separate the fats and oils in coconut oil.
The result of a chromatography experiment is shown.



- (i) Using the graph **and** the information in the table, predict the number of carbons in glyceryl trilaurate.

1

Name	Molecular formula	Melting point (°C)
Glyceryl tricaprilate	$C_{27}H_{50}O_6$	10
Glyceryl tricaprinate	$C_{33}H_{62}O_6$	31
Glyceryl trilinoleate	$C_{57}H_{98}O_6$	-5

4. (b) (continued)

(ii) Identify the compound listed in the table which is the most unsaturated. 1

(c) Edible oils such as coconut oil can be used to make emulsifiers.

(i) State how emulsifiers are made from edible oils. 1

(ii) Explain **fully** how emulsifiers prevent non-polar and polar liquids from separating into layers. 2

[Turn over

